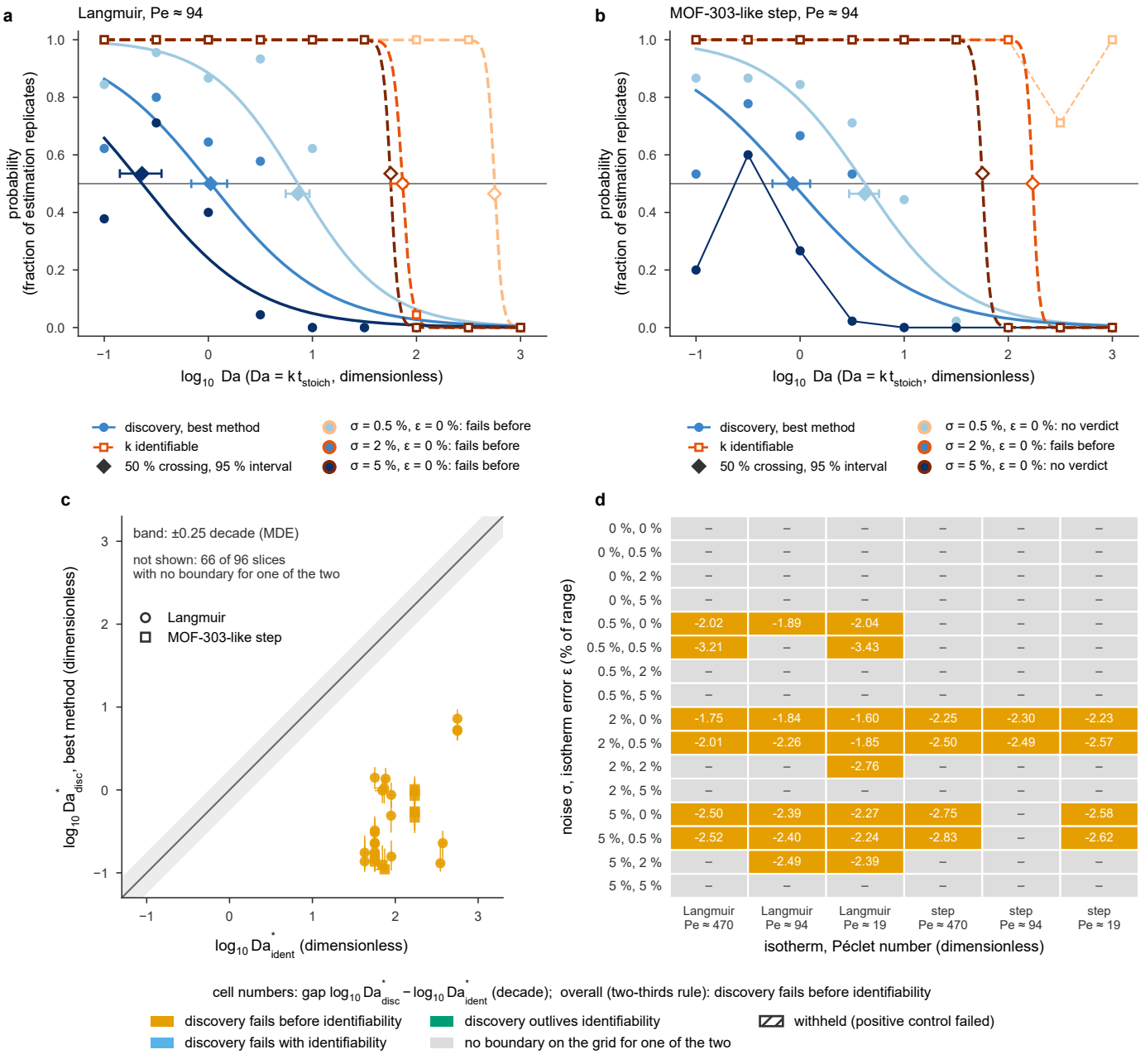

a
Langmuir, Pe ≈ 94
probability
(fraction of estimation replicates)
log10 Da (Da = k tstoich, dimensionless)
b
MOF-303-like step, Pe ≈ 94
probability
(fraction of estimation replicates)
log10 Da (Da = k tstoich, dimensionless)
discovery, best method
k identifiable
50 % crossing, 95 % interval
σ = 0.5 %, ε = 0 %: fails before
σ = 2 %, ε = 0 %: fails before
σ = 5 %, ε = 0 %: fails before
σ = 0.5 %, ε = 0 %: no verdict
σ = 2 %, ε = 0 %: fails before
σ = 5 %, ε = 0 %: no verdict
c
band: ±0.25 decade (MDE)
not shown: 66 of 96 slices
with no boundary for one of the two
Langmuir
MOF-303-like step
log10 Da*disc, best method (dimensionless)
log10 Da*ident (dimensionless)
d
noise σ, isotherm error ε (% of range)
0 %, 0 %
0 %, 0.5 %
0 %, 2 %
0 %, 5 %
0.5 %, 0 %
-2.02 -1.89 -2.04
0.5 %, 0.5 %
-3.21 -3.43
0.5 %, 2 %
0.5 %, 5 %
2 %, 0 %
-1.75 -1.84 -1.60 -2.25 -2.30 -2.23
2 %, 0.5 %
-2.01 -2.26 -1.85 -2.50 -2.49 -2.57
2 %, 2 %
-2.76
2 %, 5 %
5 %, 0 %
-2.50 -2.39 -2.27 -2.75 -2.58
5 %, 0.5 %
-2.52 -2.40 -2.24 -2.83 -2.62
5 %, 2 %
-2.49 -2.39
5 %, 5 %
Langmuir Pe ≈ 470
Langmuir Pe ≈ 94
Langmuir Pe ≈ 19
step Pe ≈ 470
step Pe ≈ 94
step Pe ≈ 19
isotherm, Péclet number (dimensionless)
cell numbers: gap log10 Da*disc − log10 Da*ident (decade); overall (two-thirds rule): discovery fails before identifiability
discovery fails before identifiability
discovery fails with identifiability
discovery outlives identifiability
no boundary on the grid for one of the two
withheld (positive control failed)